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PQ Controls INC.

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From Item No. [312S104-001] To [312S104-001]

Assembly: 312S104-001 Rev: 0

Description: 3 AXIS TKG2 CANBUS B-14875* PREMIER VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
1	B-15140	HOUSING ASSY 312, 2 AXIS NO ON-AXIS DETENT	1.0000	EA
2	C-14943	312, HOUSING, MERCURY MARINE V2	1.0000	EA
2	B-15194	BRASS INSERT, 112/212 HOUSING	1.0000	EA
2	A-09354	GREASE DOW CORNING 33	0.0000	A/R
2	B-10252	STAMPING, RETAINER, BALL M212	1.0000	EA
2	A-06936-0.50S	SCREW #6 X 1/2 PLASTITE 48-2 PAN HD SS TAP	4.0000	EA
2	B-14958	312, SHAFT WIRE STRAIN RELIEF	1.0000	EA
2	A-00600-04	WIRE TIE 4" BLACK 18 LBS PLT1M-M0	2.0000	EA
2	A-15034	ASSEMBLY PROCEDURE 312 MERCURY MARINE V2	0.0000	
1	B-15145-P	LED RING ASSY. 312, PREMIER	1.0000	EA
2	A-14901-01	ELEC ASSY., LED RING MERCURY MARINE	1.0000	EA
3	A-14901	SMT ASSY, LED BOARD (312 JOYSTICK)	1.0000	EA
4	B-14900	PCB LED Ring for Mercury Starfish Joystick	1.0000	EA
4	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond C1,2	2.0000	EA
4	SZ-15157	FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond F1,3,4,5,6,7,8	7.0000	EA
4	SR-11617-0000	1/10W 0 OHM SURFACE (0603) F2	1.0000	EA
4	SD-15069	SMD LED L955H-LOLPGQBC TRI-COLOR LED1-16, 21-36	32.0000	EA
4	SR-12545-4990	1/10W 1% 499 OHM THICK FILM (0805) R1,2	2.0000	EA
4	SR-12545-1501	1.5K OHM 1/10W 1% THICK FILM R3,4,5	3.0000	EA
4	SU-15162	TLC59116IPWR (TI) LED Lighting Driver U1,2	2.0000	EA
3	A-15059-08	ACC-437-ND 8 conductor flex cable	1.0000	EA
2	C-14949	312, CHROME SURROUND, MERCURY MARINE V2 *	1.0000	EA
2	B-14950	LIGHT PIPE, LED RING MERCURY V2, HARDCOATED *	1.0000	EA
2	A-15252	PRIMER, DOW CORNING PR-1205 *	0.0000	A/R

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2	A-15251	RTV SILICONE, DOW CORNING SYLGARD 184 ELASTOMER KIT *	0.0000	A/R
3	A-15251-A	RTV SILICONE, DOW CORNING SYLGARD 184 ELASTOMER BASE	0.0000	A/R
3	A-15251-B	RTV SILICONE, DOW CORNING SYLGARD 184 ELASTOMER CURING AGENT	0.0000	A/R
2	A-11285-0.375	SCREW #2 X 3/8" PLASTITE 48-2 SELF TAPPING	3.0000	EA
2	A-15060-08	487545-5 8 pos Connector plug	1.0000	EA
2	A-15282	ASSEMBLY PROCEDURE, 312 POTTING, MERCURY MARINE	0.0000	
1	B-15054-104	SHAFT ASSY 312 FOR TKG2	1.0000	EA
2	B-14957	SHAFT, 312 TKG2	1.0000	
2	B-10088	BALL, M212	1.0000	EA
2	A-10498-03	PIN GROOVED 5/32 X 1/2 TYPE E LOW CARBON	1.0000	EA
2	A-08113-609	LOCTITE 609 THREADLOCKER GREEN	0.0000	A/R
2	B-14945	312, X/Y AXIS ENCAPSULATED MAGNET HOLDER *	1.0000	EA
3	B-14830	MAGNET, MLX90333 SMT HALL EFFECT	1.0000	EA
2	A-00780-08SS	ROLL PIN, 1/16 X 1/2 STAINLESS STEEL	1.0000	EA
2	A-11050	ASSEMBLY PROCEDURE M212 SHAFT ASSY	1.0000	
1	B-15141	CENTERING ASSY, 312	1.0000	EA
2	C-10084	M212 CENTERING FOOT JLG#: 7020555	1.0000	EA
2	B-13471	SPRING, CENTERING LIGHT 112/212	1.0000	EA
2	B-14953	SLEEVE, 312	1.0000	EA
2	A-10500-37SS	RETAINING RING, STAINLESS STEEL	1.0000	EA
2	B-14952	BOOT MOLDING 312	1.0000	EA
2	A-00451-09	O-RING 3/8" NITRILE DURO 2-012	1.0000	EA
2	A-05447-S-316	WASHER FLAT 5/8" X 3/8", ***316 STAINLESS STEEL	1.0000	EA
2	A-15034	ASSEMBLY PROCEDURE 312 MERCURY MARINE V2	0.0000	
1	C-15142-P	GRIP ASSEMBLY TKG2 TWIST, PREMIER	1.0000	EA
2	B-14978	TOP GRIP HALF V2, CHROME PLATED	1.0000	EA
3	D-14939	TOP HALF, TWIST KNOB GRIP V2, CASTING	0.0000	EA
2	B-14955	312, PIVOT BLOCK W/ENCAPSULATED MAGNET *	1.0000	EA
3	B-14830	MAGNET, MLX90333 SMT HALL EFFECT	1.0000	EA
2	B-13472	TWIST ARMS, TWIST KNOB GRIP	2.0000	EA

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2	A-13620-01	BEARING, OIL FILLED .378 ID X .469 OD X .304 H *	1.0000	EA
2	B-13491-SS	SPRING, EXTENSION, TWIST KNOB STAINLESS STEEL	1.0000	EA
2	B-14979	BOTTOM GRIP HALF V2, CHROME PLATED	1.0000	EA
3	C-14940	BOTTOM HALF, TWIST KNOB GRIP V2, CASTING	0.0000	EA
2	B-14951	GRIP SLEEVE TWIST KNOB V2	1.0000	
2	A-13890	ADHESIVE, DOUBLE/BUBBLE URETHANE	0.0000	A/R
2	A-00703-03	SCREW 2-56 X 3/16 PHIL PAN MS STAINLESS STEEL	3.0000	EA
2	A-00161-02S	LOCK WASHER #2 INTERNAL TOOTH STAINLESS STEEL	3.0000	EA
2	A-10498-04	PIN GROOVED 5/32 X 5/8 TYPE E LOW CARBON	1.0000	EA
2	A-15002-06S	4-40 X 3/4" BUTTON HEAD SOCKET CAP SS	2.0000	EA
2	A-00451-02	O-RING, TWIST KNOB GRIP V2, ID = 1.669	1.0000	EA
2	A-01349-13	ROLL PIN 1/8 X 1" ZINC YELLOW	1.0000	EA
2	A-02686-03S	RING, RETAINING 3/8" SS	1.0000	EA
2	B-15082	MERCURY MARINE JOYSTICK V2, KNOB DISPLAY ASSY., PREMIER *	1.0000	EA
3	A-15783	KNOB DISPLAY APPEARANCE GUIDE	0.0000	
2	A-15080-01	ELEC ASSY, MERCURY V2 KNOB	1.0000	EA
3	A-14851	SMT ASSY, KNOB BOARD (312 JOYSTICK)	1.0000	EA
4	B-14850	PCB Knob PCB for Mercury Starfish Joystick	1.0000	EA
4	SC-10228-01	TAJA106K010RNJ CAPACITOR TANT 10UF 10V SURFACE MOUNT C3	1.0000	EA
4	SC-14635-02	0.01uf 50V 0603 C4,8,10,12,17	5.0000	EA
4	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond C5,7,9,11,13,16,18	7.0000	EA
4	SC-12564-05	4.7UF, 16V CAP CERAMIC (0805) C6	1.0000	EA
4	SC-14907-01	18pF 16V 5% CERAMIC CAPACITOR, 0603 Bond C14,15	2.0000	EA
4	SD-15154	APT1608SECK/J4-PRV SMD LED, ORANGE 605nm (0603 PACKAGE) LED1,2,3,4,5,10,11,12,13,14,15	11.0000	EA
4	SD-15158	SMD LED, BLUE 470nm (1206 PACKAGE) APTD3216QBC/D LED6,7,8,9,16	5.0000	EA
4	SZ-15157	FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond	1.0000	EA

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4	SR-11617-1400	FER1 1/10W 1% 140 OHM SURFACE MOUNT RESISTORS (0603 PACKAGE) R8,9,10,11,12,17,18,19,20,21,22	11.0000	EA
4	SR-11617-86.60	1/10W 1% 86.6 OHM SURFACE MOUNT RESISTORS (0603 PACKAGE) R13,14,15,16,26	5.0000	EA
4	SR-11617-0000	1/10W 0 OHM SURFACE (0603) R23,24,25	3.0000	EA
4	SQ-14908	MOSFET N-CHANNEL 60V, BS170FTA (Diodes Inc.) Q11	1.0000	EA
4	SQ-15155	TRANSISTOR, NPN 100mA 50V MUN2211T1G Q2,3,4,5,6	5.0000	EA
4	SR-12545-1002	NRC10F1002TRF 10K OHM 1/10W 1% THICK FILM (0805) R1,2,7	3.0000	EA
4	SR-12552-1200	1/10W 5% 120 OHM THICK FILM 0805 R3	1.0000	EA
4	SR-11617-1001	1/10W 1% 1K OHM SURFACE R4	1.0000	EA
4	SR-12545-1501	1.5K OHM 1/10W 1% THICK FILM R6	1.0000	EA
4	SU-15156	32 BIT ARM MICROCONTROLLER, 72MHz (LQFP-48) STM32F103C8T6 U1	1.0000	EA
4	SU-12791-3.3BS	MIC5209-3.3YSTR 500MA LO-NOISE LDO REG U3	1.0000	EA
4	SU-14405	SN65HVDA1050AQDRQ1 CAN TRANSCEIVER W/HIGH SPEED & EMC U4	1.0000	EA
4	SY-14550-08	ABM3-8.000MHZ-D2Y-T 8MHz 18pF miniature SMT Crystal Y1	1.0000	EA
3	A-14857	SMT ASSY, KNOB SENSOR BOARD (312 JOYSTICK)	1.0000	EA
4	B-14856	PCB Knob Sensor PCB for Mercury Startfish Joystick	1.0000	EA
4	SU-15161	MLX90333LGO-BCH-100, 3D POSITION HALL SENSOR, DUAL DIE, SPI	1.0000	EA
4	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond	4.0000	EA
3	A-15062	'850-10-007-20-001101 PRECI-DIP 7 pos interconnect pins	1.0000	EA
3	A-12827-01-RD	RED WIRE, 26GA TEFLON TYPE E 19/38	0.5833	FT
3	A-12827-01-BK	BLACK WIRE 26GA TEFLON TYPE E 19/38	0.5833	FT
3	A-12827-01-BR	BROWN WIRE 26GA TEFLON TYPE E 19/38	0.5833	FT

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3	A-12827-01-GY	GRAY WIRE, 26GA TEFLON TYPE E 19/38	0.5833	FT
3	A-15578	ADHESIVE, HOT MELT 3779	0.0000	A/R
2	A-15193	GREASE, CMD EXTREME PRESSURE LUBE #3	0.0000	A/R
1	A-14881-01	ELEC ASSY, MERCURY V2 MOTHERBOARD	1.0000	EA
2	A-14881	SMT ASSY, MOTHERBOARD BOARD (312 JOYSTICK)	1.0000	EA
3	B-14880	X/Y PCB for Mercury Starfish Joystick	1.0000	EA
3	SC-09917-01	CAPACITOR .01UF 200V CERAMIC Bond (1210) C1,59	2.0000	EA
3	SC-09917-05	CERAMIC CAPACITOR 1uF 25V 10% X7R (1210) C5	1.0000	EA
3	SC-10228-01	TAJA106K010RNJ CAPACITOR TANT 10UF 10V SURFACE MOUNT C10	1.0000	EA
3	SC-14635-01	GRM1885C1H111JA01D 110pF 16V 5% CERAMIC CAP 0603 C11,12,13,14,33,34,35,36	8.0000	EA
3	SC-14635-02	0.01uf 50V 0603 C15,18,19,20,21,22,37	7.0000	EA
3	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond C16,17,24,25,26,27,28,29,30,31,32,39,40,45,48	15.0000	EA
3	SC-12564-05	4.7UF, 16V CAP CERAMIC (0805) C23	1.0000	EA
3	SC-14907-01	18pF 16V 5% CERAMIC CAPACITOR, 0603 Bond C41,42	2.0000	EA
3	SC-12564-10	22uF, 20% 35V CAP CERAMIC (0805) C46	1.0000	EA
3	SC-12564-11	C2012X5R1V156M125AC 15uF, 20% 35V CAP C47,49,50,54,52,53	6.0000	EA
3	SC-12564-07	0.1UF, 50V, CAP CERAMIC (0805) C54	1.0000	EA
3	SC-14635-06	.0082uF 16V 10% CERAMIC CAP 0603 Bond C55	1.0000	EA
3	SC-09160-02	CL31A106KOHNNNE 10UF, 16V CAPACITOR, CERAMIC 10% (1206) C56,57	2.0000	EA
3	SC-09917-02	CAPACITOR 0.1UF 200V CERAMIC SMT (1210) C58,60,61	3.0000	EA
3	SD-15271	20V TVS DIODE BI-DIR. SMCJ20CA	2.0000	EA

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3	SD-15272	D1,3 B560C-13-F SCHOTTKY DIODE 60V 5A Bond	1.0000	EA
3	SD-12845	D2 B250-13-F 2A SCHOTTKY DIODE	1.0000	EA
3	SD-15273	D4 NUP2105LT1G TVS DIODE ARRAY 27V CANBUS PROTECTION ##	2.0000	EA
3	SD-15274	D5,6 TVS DIODE 5Vr 400W UNI-DIR. (LITTELFUSE #SMAJ5.0A) ##	3.0000	EA
3	SD-15275	D11,12,13 TVS DIODE ESD PROTECTION 5V SOD523 Bond ##	9.0000	EA
3	SZ-15157	D7,8,9,10,14,15,16,17,18 FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond	12.0000	EA
3	SZ-15276	FER1,2,3,4,5,6,7,8,9,11,12,13 FERRITE BEAD 600 OHM 2A (0805) TDK	2.0000	EA
3	SL-15277	FER14,15 CHOKE COM MODE 1mH 0.8A Bond	1.0000	EA
3	SL-14637	L1 CHOKE COM MODE 2x51uH 0.3A 744212510	2.0000	EA
3	SL-15278-01	L2,3 INDUCTOR 27uH 30% SMD 6038 Bond	1.0000	EA
3	SL-15278-02	L4 CDRH6D38NP-7R4NC INDUCTOR 7.4uH 30% SMD 6038	1.0000	EA
3	SQ-14908	L5 MOSFET N-CHANNEL 60V, BS170FTA (Diodes Inc.)	2.0000	EA
3	SQ-15155	Q1,2 TRANSISTOR, NPN 100mA 50V MUN2211T1G	3.0000	EA
3	SQ-15166	Q4,5,6 TRANSISTOR, PNP 50V DPLS350E-13	3.0000	EA
3	SR-12545-1200	Q7,8,9 120 OHM 1/10W 1% THICK FILM 0805	1.0000	EA
3	SR-12545-1001	R1 1K OHM 1/10W 1% THICK FILM	5.0000	EA
3	SR-12545-1002	R12,25,37,39,41 NRC10F1002TRF 10K OHM 1/10W 1% THICK FILM (0805)	3.0000	EA
3	SR-12545-1501	R5,27,28 1.5K OHM 1/10W 1% THICK FILM	9.0000	EA

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3	SR-09152-3012	R6,7,8,9,10,11,21,26,29 1/4W 1% 30.1K OHM SURFACE MOUN R31	1.0000	EA
3	SR-12545-3320	Resistor 332 ohm 1% 1/4W 0805 R32,33,38,40,42	5.0000	EA
3	SR-09152-2003	1/4W 1% 200K OHM SURFACE MT R36	1.0000	EA
3	SR-12545-0000	0 OHM 1/10W 1% THICK FILM (0805) R46,47	2.0000	EA
3	SR-09152-01.10	1/4W 1% 1.1 OHM SMT R48	1.0000	EA
3	SR-15279-2002	1/4W 1% 20K OHM SMT 0805 PACKAGE R49	1.0000	EA
3	SR-15281-6802	1/2W 1% 68K OHM ANTI SURGE SMT 0805 PACKAGE R50	1.0000	EA
3	SR-15281-1202	1/2W 1% 12K OHM ANTI SURGE SMT 0805 PACKAGE R51	1.0000	EA
3	SU-15169	32 BIT ARM MICROCONTROLLER, 72MHz (LQFP-64) STM32F105R8T6 U1	1.0000	EA
3	SU-15280	REGULATOR, SWITCHING 40V 2A U2	1.0000	EA
3	SU-12791-3.3BS	MIC5209-3.3YSTR 500MA LO-NOISE LDO REG U4	1.0000	EA
3	SU-14405	SN65HVD1050AQDRQ1 CAN TRANSCEIVER W/HIGH SPEED & EMC U5,8	2.0000	EA
3	SU-15161	MLX90333LGO-BCH-100, 3D POSITION HALL SENSOR, DUAL DIE, SPI U7	1.0000	EA
3	SY-14550-08	ABM3-8.000MHZ-D2Y-T 8MHz 18pF miniature SMT Crystal Y1	1.0000	EA
2	A-15055-04	5-103669-3 4 pos Connector Header	1.0000	EA
2	A-15058-08	1-104071-0 8 pos Connector Header for Mercury x/y board	1.0000	EA
2	A-15058-05	5-104071-2 5 pos connector header	1.0000	EA
2	A-06977-36	HEADER BERG 36 POS 68000-436 STRAIGHT	0.0000	EA
2	A-07794-01	CONFORMAL COATING #MS-470C SPRAY	0.0000	A/R
2	A-15083	PRETEST PROCEDURE, 312 MERCURY MARINE	0.0000	

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2	A-15185	SOFTWARE FLASHING PROCEDURE, 312 MERCURY MARINE	0.0000	
1	B-15144	ENCLOSURE ASSY. 312	1.0000	EA
2	C-14944	312, ENCLOSURE	1.0000	EA
2	A-00451-13	O-RING, 312 JOYSTICK	1.0000	EA
2	A-06936-0.87S	SCREW #6 X 7/8 PLASTITE 48-2 PAN HD SS	4.0000	EA
2	A-13718-01	FLAT RUBBER WASHER .140 X .312 X 1/32	4.0000	EA
2	A-03485-01	LIQUID TIGHT STRAIN RELIEF HEYCO 3207 SM	1.0000	EA
2	B-14956	312 MOUNTING BRACKET W/STUDS ASSY, MERCURY MARINE	1.0000	EA
2	A-01261-04S	LOCK NUT NYLON 4-40 STAINLESS STEEL	3.0000	EA
2	A-15306	HYDROPHOBIC VENT PATCH, LTI ATLANTA #074-1000-000	1.0000	EA
1	B-15081	KEYPAD ASSY., MERCURY MARINE STARFISH JOYSTICK *	1.0000	EA
2	A-16228	KEYPAD PRETEST PROCEDURE, 312 MERCURY MARINE	0.0000	
1	A-11285-0.25S	#2 X 1/4" PLASTITE SELF TAPPING 48-2 SS	2.0000	EA
1	A-12312-06B	WASHER, NYLON, .112 ID, .206 OD .053 THK BLACK *	2.0000	EA
1	A-11285-0.375S	SCREW #2 X 3/8" PLASTITE 48-2 SS	2.0000	EA
1	A-14954	8M0111425 NEMA cable with orange heat shrink	1.0000	EA
1	A-15056-04	104257-3 4 pos Connector plug	1.0000	EA
1	A-15057	SOCKET, REEL MFG # 1-104480-2, Sockets for 104257-3 p	4.0000	EA
1	B-15079-01	312 BEZEL ASSY., PREMIER, MERCURY MARINE V2	1.0000	EA
2	C-14946	312 BEZEL, UPPER (PREMIER), MERCURY MARINE V2	1.0000	EA
2	B-14948	312, CHROME BEZEL TRIM, MERCURY MARINE V2 *	1.0000	EA
2	C-14947	312 BEZEL, LOWER, MERCURY MARINE V2	1.0000	EA
2	A-15004	DECAL, MERCURY JEWEL P/N 37-8M6001063	1.0000	EA
1	B-14941	GASKET 312, EPDM SPONGE	1.0000	EA
1	A-15230	# 10-6957 SCREW #10-16 X 3/4 HI-LO RECESSED PAN HD	4.0000	EA
1	A-15034	ASSEMBLY PROCEDURE 312 MERCURY MARINE V2	0.0000	
1	A-15078	FINAL TEST PROCEDURE, MERCURY MARINE 312S104-001	0.0000	
1	S1143	SOFTWARE, MERCURY MARINE 312S104-001 PREMIER & 312S105-005 J	0.0000	

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